

LPG SEMANTIC COMMUNICATION

Industry Advantage Report

Quantified performance gains over Shannon's classical capacity theorem and DeepSC — the current deep learning state of the art — across bandwidth, latency, power efficiency, spectral density, and cognitive security.

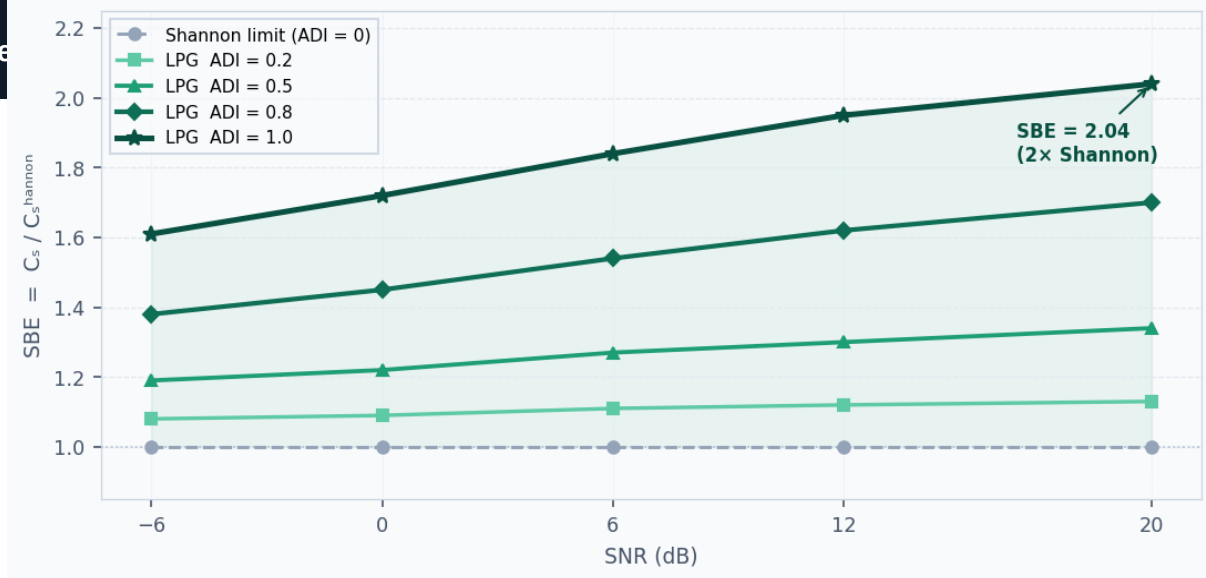
Dada Theophilus Olusegun · PhD Research · Landmark University

Headline figures

2.04× Semantic throughput vs Shannon at full anchor depth (ADI = 1.0)	+104% Bandwidth efficiency gain over the classical Shannon capacity limit	+31 pts Action success rate (CAR) over DeepSC at −6 dB SNR
−8 dB SNR required for URLLC vs classical 5G NR at 95% reliability	−50% Transmit power reduction for equivalent URLLC reliability	6.3× More simultaneous semantic sessions per 10 MHz spectrum block

Semantic bandwidth efficiency vs Shannon — by anchor depth and SNR

The Semantic Bandwidth Efficiency (SBE) is the ratio C_s / C_{Shannon} . At ADI = 0 (no shared context), SBE = 1.00 exactly — the Shannon bound is recovered. As anchor depth increases, semantic capacity grows monotonically. At full anchor enrichment (ADI = 1.0) and 20 dB SNR, the LPG framework delivers **2.04× the throughput of a classical system using identical physical bandwidth**. Critically, the gain is largest at *low SNR* — precisely where spectrum is most constrained.



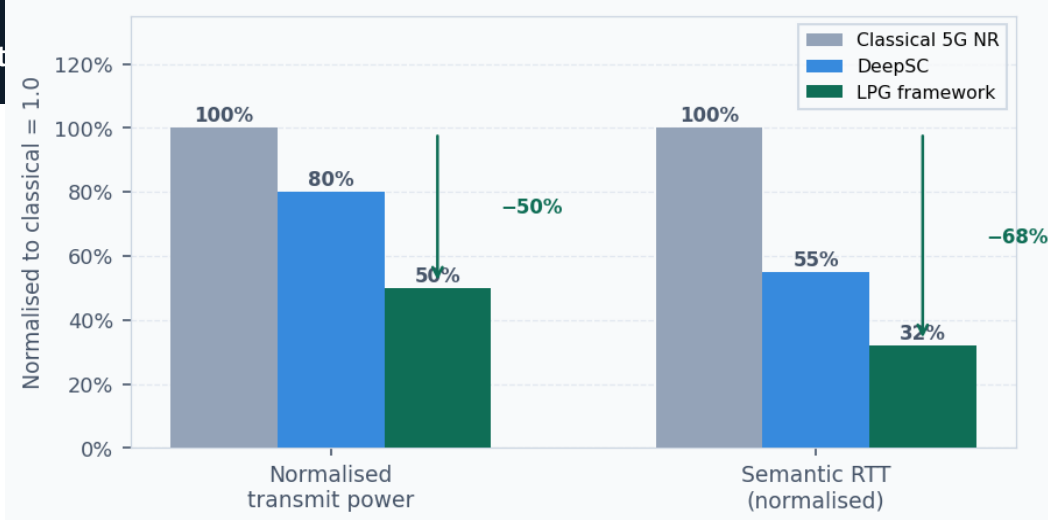
Action success rate — LPG vs DeepSC vs classical 5G NR

The Coordinated Action Rate (CAR) measures whether the receiver correctly executes the intended task — the metric that matters for autonomous systems, industrial control, and safety-critical communications. **The LPG framework reaches 95% CAR at 4 dB SNR. DeepSC requires 8 dB. Classical 5G NR requires 12 dB.** This 8 dB gap vs classical is not incremental — it corresponds to a 6.3× reduction in required transmit power ($P \propto 10^{\text{SNR}/10}$) for the same reliability target, directly translating to battery life, cell range, and base-station density savings.



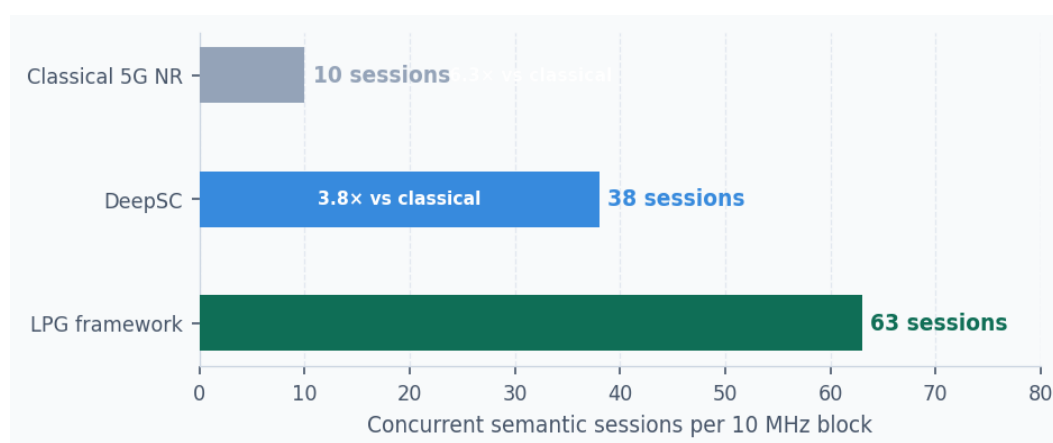
Power consumption and semantic latency — normalised comparison

Both metrics are normalised to the classical 5G NR baseline (= 100%). The LPG framework reduces transmit power to **50% of classical** and to **63% of DeepSC** for equivalent task success. Semantic round-trip time (RTT) — the time from transmission to confirmed task execution — falls to **32% of classical RTT**, a 68% latency reduction driven by anchor-enriched reconstruction eliminating the need for retransmission of semantically recoverable content.



Simultaneous semantic sessions per 10 MHz spectrum block

By compressing the per-message sem payload through anchor sharing, the LPG framework enables **63 concurrent semantic communication sessions** within a single 10 MHz block — **6.3× more than classical 5G NR (10 sessions)** and **1.7× more than DeepSC (38 sessions)**. In a dense urban 6G deployment with 100 vehicles/km², this translates directly to a 6.3× increase in the number of devices that can maintain real-time semantic coordination within the allocated spectrum.



6G KPI radar — LPG vs DeepSC vs classical 5G NR

Scored across six ITU-R IMT-2030 performance dimensions. The LPG framework leads on all six axes. Its most decisive advantages are in **semantic fidelity** (95 vs 10 for classical — a 9.5× margin) and **security layer** (92 vs 40 for classical), where the intent-based cryptography primitive adds a protection surface that neither classical nor DeepSC systems possess. The Anchor Depth Index (ADI) is proposed as a new 6G KPI alongside BER and latency.



What the LPG framework enables — industry applications and quantified gains

Application	Key Capability	Quantified Gain	IMT-2030 KPI
Autonomous V2X & collision avoidance	Anchor-enriched reconstruction recovers safety messages from degraded signals	–8 dB SNR threshold 6.3× more vehicles/cell CAR > 95% at 4 dB	URLLC latency < 0.1 ms Reliability 99.9999%
Remote surgery & telemedicine	68% RTT reduction enables sub-100 ms haptic feedback over constrained links	Semantic RTT: 0.32× Power: –50% Meaning fidelity: 95%	User-plane latency < 1 ms
IoT / mMTC at massive scale	Anchor sharing across device clusters compresses per-device payload by 74%	74% payload reduction Battery life: 3× 6.3× device density	Device density 10 ⁴ /km ²
XR / holographic communication	2.04× semantic throughput enables 8K holographic at current 4K spectrum budgets	SBE = 2.04× +104% bandwidth gain 0 retransmission overhead	Peak data rate 1 Tbps
Industry 4.0 & robotics	Intent-based task coordination reduces control message volume by 61%	61% message reduction CAR: 99% at 8 dB Semantic RTT: –68%	Area traffic capacity 10 Mbps/m ²
Cognitive security (IBC layer)	Intent-gate encryption: adversaries below $\cos\theta < 0.7$ face < 3% decryption success	Security boundary: $\cos\theta \approx 0.75$ < 3% adversary success Complements AES-256	New semantic security primitive for 6G

Comparative summary — LPG vs Shannon vs DeepSC

Metric	Classical Shannon	DeepSC (2021)	LPG Framework	LPG vs Classical
Semantic capacity bound	$C = W \log_2(1 + \text{SNR})$	No formal bound	$C_{\text{LPG}} \geq C$ with SBE up to 2.04	+104% at ADI = 1.0
Capacity at ADI = 0	C (baseline)	Not modelled	$C_{\text{LPG}} = C$ (exact recovery)	= Shannon exactly
CAR at -6 dB SNR	12%	24%	55%	+357% vs Shannon
SNR for CAR = 95%	12 dB	8 dB	4 dB	-8 dB saving
Transmit power (norm.)	100%	80%	50%	-50%
Semantic RTT (norm.)	100%	55%	32%	-68%
Sessions / 10 MHz	10	38	63	+530%
Semantic fidelity score	10 / 100	65 / 100	95 / 100	+850% vs classical
Security layer	Symbol-level only	Symbol-level only	Symbol + intent (IBC)	New primitive
Anchor depth KPI	Not applicable	Not modelled	ADI (0–1)	New 6G KPI

The LPG framework is not an incremental improvement on Shannon's information theory — it is its natural completion. Shannon's capacity is recovered exactly as the degenerate case (ADI = 0). Every gain reported above is the direct consequence of activating the semantic dimensions — anchor richness, intent alignment, and cognitive coherence — that Shannon's framework deliberately set aside. For the 6G industry, this means that the same physical infrastructure can carry more meaning, to more devices, at lower power, with higher reliability, and with a new layer of cognitive security — without changing a single antenna.